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Unit 1 - AT-3

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Test

CSA-1704

(1) Smart Home Automation System and intelligent Agents.

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Introduction

A Smart home Automation system controls lighting, temperature and security based on user behaviour and environmental conditional. it recives real-time inputs from sensor detectors temperatures sensor, cameras, clock and perform action automatically to make Artificially intelligence agent ~~ab~~ enable the system intelligence decisions

Types of Intelligents Agents

(1) Simple Reflex agents
(2) Model - based Reflex agent
(3) Utility Based Reflex agent
(4) Goal Based Agent

→ Simple Reflex agents

A simple Reflex agent make decision based only on the current input used predefined condition - action rule

Working

- if Motion is detected → Turn on light
- if temperature is >30°C → Turn on air

- If temperature < 30°c turn off air

Advantages

- Fast response
- Easy to implement
- Low computation cost

Disadvantages

- Cannot remember previous event
- Cannot leave user preferences

② **Model-Based Reflex agent**

It is a maintains an internal model of the environment. It stores previous states and updat a new information arriving

working

- Remember whether someone is inside the room
- Detect that a window is open while AC is running.

③ **Goal Based agent**

It is chooses action that help achieve a specific goal

Goal in smart Home

- maintain room temperature 24°C
- ensure all doors are locked at night.
- Reduce unneccessary electrical usage

④ **Utility Based Reflex agent**

It a utility Based Reflex agent select the action that provides highest benefits

working

1. The temperature ~~is Comes on same~~ phase
2. ~~The~~

working

- Keep on AC is on mode
- comfortable
- High electricity usage

Factors

- User Security
- Comfort
- Cost saving

(2) A robot is placed in an unknown maze and must find a path the exit without prior knowledge of the layout.

sol The robot placed in an unknown maze must discover a path to exit without prior knowledge of the environment. search algorithm help to robot explore maze efficently.

uniformed search strategies

1. Uniform cost search (UCS)

Uniform cost search expand the Node with the lowest cumaltive path cost

working

1. Start from the initial poistion
2. expand the least -cost node
3. continoue untile the exit is found

Example

if moving cost

- left 2
- right 1
- forward = 3

UCS always select the path with smallest total cost

Advantages

- Find the optimal path
- works with different movement cost
- complete algorithm

Disadvantage

- uses large memory
- can be slow for large mazes

complexity

- Time complexity : $O(b^{(1+\lfloor C^*/\epsilon\rfloor)})$
- Space complexity : $O(b^{(1+\lfloor C^*/\epsilon\rfloor)})$

where b = branching factor

C^* = optimal solution

ϵ = minimum step cost

⑤ **iterative Deepening Search** (IDS)

IDS repeatedly perform depth limited searches with increasing depth limits until goal is found

working

Depth 0

Start

Depth 1

Start – A

Depth – 2

start – A – B

continue until exit is reached

Advantage

- Low memory usage
- Complete for finite search space
- Suitable when solution depth

disadvantages

- Repeats Node expansions
- not optimal movement cost differ

Most effective Combination

- Goal Based Agent + model Based agent + uniform cost search (VG)